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Program: Btech CSE B16

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HDOC Day-21 — Menu-Driven C Program

1. Algorithm

Step 1: Start.

Step 2: Declare integer variables `m`, `n`, `choice`, `i`, and `j`.

Step 3: Input two positive integers `m` and `n`.

Step 4: Check whether:

- `m > 0`
- `n > 0`
- `m <= n`

If the condition is false, display "**Invalid input**" and terminate the program.

Step 5: Display the menu:

1. Multiplication tables of `m` and `n`
2. Multiplication grid
3. Consecutive integers
4. Exit

Step 6: Read the user's choice.

Step 7: Use a `switch` statement:

Case 1 — Multiplication Tables

- Run a loop from `i = 1` to `10`.
- Print `m × i` and `n × i` side by side.

Case 2 — Multiplication Grid

- Print column headings from 1 to n .
- Use an outer loop from $i = 1$ to m .
- Use an inner loop from $j = 1$ to n .
- Print $i * j$.

Case 3 — Consecutive Integers

- Run an outer loop from $i = m$ to n .
- For each i , run another loop from $j = 0$ to 5.
- Print $i + j$.

Case 4 — Exit

- Display an exit message.
- Terminate the loop.

Default

- Display "Invalid choice! Please enter 1 to 4."

Step 8: Repeat Steps 5–7 until the user selects 4.

Step 9: Stop.

2. Test-Case Table

Test Case	Input m, n	Choice	Expected Output	Actual Output	Result
1	3, 5	1	Tables of 3 and 5 from $\times 1$ to $\times 10$	Tables printed correctly	Pass
2	3, 5	2	3×5 multiplication grid	Grid printed correctly	Pass
3	3, 5	3	Six consecutive integers for 3, 4 and 5	Values printed correctly	Pass
4	3, 5	4	Program exits	Program exited	Pass
5	3, 5	7	Invalid-choice error message	Error message displayed	Pass
6	5, 3	—	Invalid input because $m > n$	Error message displayed	Pass
7	-2, 5	—	Invalid input because values must be positive	Error message displayed	Pass

3. C Program

```
#include <stdio.h>

int main()
{
    int m, n, choice, i, j;

    printf("Enter two positive integers m and n (m <= n): ");
    scanf("%d %d", &m, &n);

    if (m <= 0 || n <= 0 || m > n)
    {
        printf("Invalid input! Enter positive integers with m <= n.\n");
        return 0;
    }

    do
    {
        printf("\n1. Multiplication tables of m and n");
        printf("\n2. Multiplication grid");
        printf("\n3. Consecutive integers");
        printf("\n4. Exit");
        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                printf("\nMultiplication Tables of %d and %d (m=%d, n=%d)\n",
                    m, n, m, n);

                for (i = 1; i <= 10; i++)
                {
                    printf("%d x %d = %-5d\t%d x %d = %d\n",
                        m, i, m * i,
                        n, i, n * i);
                }
                break;

            case 2:
                printf("\nMultiplication Grid (m=%d, n=%d)\n\n", m, n);

                printf("      ");
                for (j = 1; j <= n; j++)
                {
                    printf("%4d", j);
                }

                printf("\n      ");
                for (j = 1; j <= n; j++)
                {
                    printf("----");
                }
                printf("\n");
            
```

```

        for (i = 1; i <= m; i++)
        {
            printf("%2d | ", i);

            for (j = 1; j <= n; j++)
            {
                printf("%4d", i * j);
            }

            printf("\n");
        }
        break;

    case 3:
        printf("\nConsecutive Integers from %d to %d (m=%d, n=%d)\n",
            m, n, m, n);

        for (i = m; i <= n; i++)
        {
            printf("%d: ", i);

            for (j = 0; j <= 5; j++)
            {
                printf("%d ", i + j);
            }

            printf("\n");
        }
        break;

    case 4:
        printf("\nExiting the program...\n");
        break;

    default:
        printf("\nInvalid choice! Please enter a number from 1 to
4.\n");
    }

    } while (choice != 4);

    return 0;
}

```

5. Compilation , Execution and Evaluation Against Test Cases

The program successfully satisfies all the required conditions:

- **Test Case 1:** Choice 1 correctly prints the multiplication tables of m and n side by side → **PASS**

```
(kali@kali)-[~/Desktop]
$ cc day21.c

(kali@kali)-[~/Desktop]
$ ./a.out
Enter two positive integers m and n ( $m \leq n$ ): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1

Multiplication Tables of 3 and 5 ( $m=3, n=5$ )
3 x 1 = 3      5 x 1 = 5
3 x 2 = 6      5 x 2 = 10
3 x 3 = 9      5 x 3 = 15
3 x 4 = 12     5 x 4 = 20
3 x 5 = 15     5 x 5 = 25
3 x 6 = 18     5 x 6 = 30
3 x 7 = 21     5 x 7 = 35
3 x 8 = 24     5 x 8 = 40
3 x 9 = 27     5 x 9 = 45
3 x 10 = 30    5 x 10 = 50
```

- **Test Case 2:** Choice 2 correctly generates an $m \times n$ multiplication grid → **PASS**

```
Enter your choice: 2

Multiplication Grid ( $m=3, n=5$ )

      1  2  3  4  5
    -----
1 |    1  2  3  4  5
2 |    2  4  6  8 10
3 |    3  6  9 12 15
```

- **Test Case 3:** Choice 3 correctly prints each integer from m to n followed by its next five integers → **PASS**

```
Enter your choice: 3

Consecutive Integers from 3 to 5 (m=3, n=5)
3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10
```

- **Test Case 4:** Choice 4 correctly terminates the program → **PASS**

```
Enter your choice: 4

Exiting the program...
```

- **Test Case 5:** An invalid menu choice correctly produces an error message → **PASS**

```
Enter your choice: 7

Invalid choice! Please enter a number from 1 to 4.
```

- **Test Cases 6–7:** Invalid values of m and n are correctly rejected → **PASS**

```
Enter two positive integers m and n ( $m \leq n$ ): 5 3
Invalid input! Enter positive integers with  $m \leq n$ .
```

```
Enter two positive integers m and n ( $m \leq n$ ): -2,5
Invalid input! Enter positive integers with  $m \leq n$ .
```

Final Result: All prepared test cases passed successfully.